

Technical Appendix Outline

1. Pipeline Diagram

flowchart LR

A["UCSD Goodreads review JSON"] --> B["Sampling by rating and length"]
B --> C["Cleaning and deduplication"]
C --> D["Locked holdout: 1,000 reviews"]
C --> E["Train/test: 15,000 reviews"]
D --> F["3 independent human labels"]
F --> G["Reliability + consensus"]
D --> H["6-model genAI benchmark"]
H --> I["Best genAI labeling strategy"]
I --> E
E --> J["Fine-tuned specialist model"]
G --> K["Final holdout evaluation"]
H --> K
J --> K
K --> L["Recommendation: deploy strategy + governance"]

2. Label Codebook

Use docs/labeling_codebook.md and generated reports/labeling_codebook.pdf.

3. Benchmark Table

Supplier	Model	Macro F1	MCC	Alpha vs Human	Runtime	Cost	Parse Failures
OpenAI	gpt-4o-mini	TBD	TBD	TBD	TBD	TBD	TBD
OpenAI	gpt-4.1-mini	TBD	TBD	TBD	TBD	TBD	TBD
Anthropic	claude-3-5-haiku-latest	TBD	TBD	TBD	TBD	TBD	TBD
Anthropic	claude-3-7-sonnet-latest	TBD	TBD	TBD	TBD	TBD	TBD
Google	gemini-2.0-flash	TBD	TBD	TBD	TBD	TBD	TBD
Google	gemini-2.5-pro	TBD	TBD	TBD	TBD	TBD	TBD

4. Fine-Tuning Experiments

Experiment	Task	Model	Learning Rate	Epochs	Macro F1	MCC / Sample F1
E1	emotions	distilroberta-base	2e-5	3	TBD	TBD
E2	emotions	roberta-base	2e-5	3	TBD	TBD
E3	commitment	roberta-base	2e-5	3	TBD	TBD
E4	recommendation	roberta-base	2e-5	3	TBD	TBD

5. Error Analysis Plan

Analyze false positives and false negatives by:

- Rating level.
- Review length bucket.
- Emotion rarity.
- Mixed-positive/negative reviews.
- Sarcasm and irony.
- Plot-content versus reader-feeling confusion.
- Conditional recommendations.
- Series versus standalone books.

6. Mitigation Plan

Likely mitigations:

- Add coder examples for high-confusion labels.
- Tune multi-label emotion thresholds on validation data.
- Use class weights for rare labels.
- Add a human review queue for low-confidence rare emotions.
- Use a genAI escalation path for ambiguous commitment/recommendation cases.